EECE 2560: Fundamentals of Engineering Algorithms
Department of Electrical and Computer Engineering

Project Mastermind

Part b

In the second part of the project, complete the program that allows a user (codebreaker) to play Mastermind against the computer (codemaker).

1. Implement the class `response` which stores the response to a guess (number correct and number incorrect), and which includes:
   (a) a constructor,
   (b) functions to set and get the individual stored values within a response,
   (c) an overloaded operator == that compares responses and returns true if they are equal (global),
   (d) an overloaded operator << that prints a response (global).
2. Implement the class `mastermind` which handles the playing of the game for a player, and which includes:
   (a) a `secrete code` object as a data member,
   (b) two constructors to initialize the game: one constructor is passed values of n and m that were read from the keyboard, and the other constructor is passed no parameters and uses default values for $n = 5$ and $m = 10$,
   (c) a function that prints the secret code,
   (d) a function `humanGuess()` that reads a guess from the keyboard and returns a code object that represents the guess code,
   (e) a function `getResponse()` that is passed one code object (a guess code), and returns a response object,
   (f) a function `isSolved()` that is passed a response object and returns true if the response indicates that the guess code matches the secrete code.
   (g) a function `playGame()` that initializes a random secrete code, prints it to the screen, and then iteratively gets a guess from the player and prints the response until either the codemaker or the codebreaker has won.
3. Implement a function `main()` which initializes a `mastermind` object and then calls `playGame()`. The version of the code you submit should print the secret code to the screen to help us test your code.
4. You should check and validate the inputs, e.g., the range of the digits and the number of digits.